

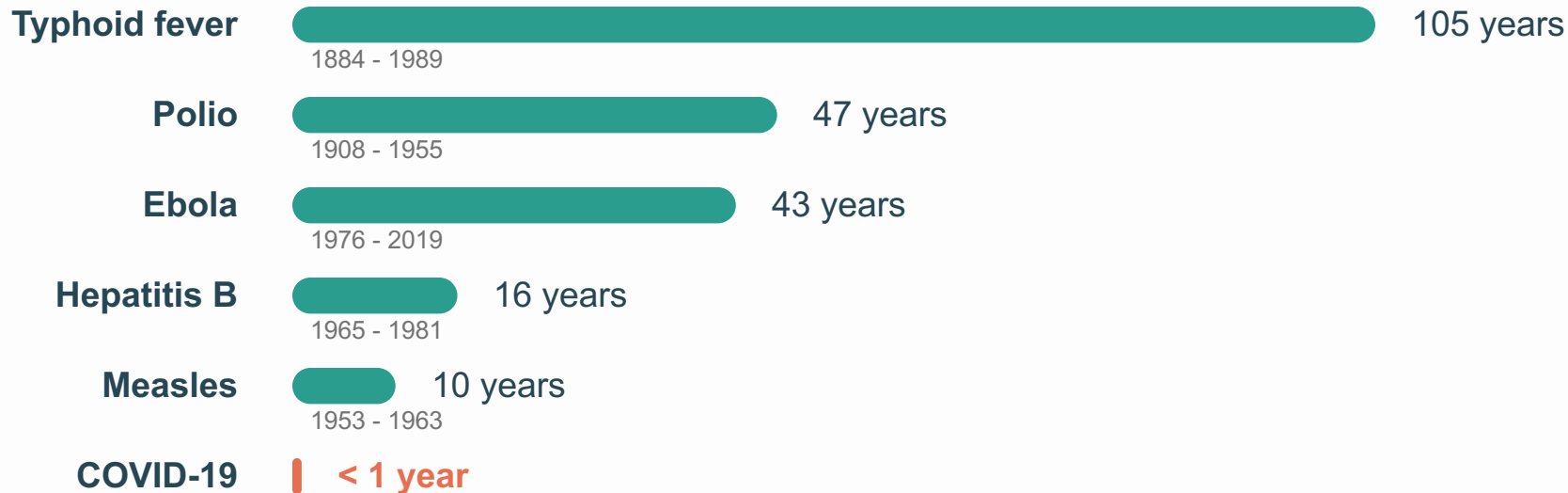
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

1796–2020: A Data-Driven Overview of Vaccine
Impact

Source: Our World in Data

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

The gap between pathogen identification and vaccine licensure has dramatically collapsed, driven by modern tools like mRNA technology and genomic sequencing.



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Data from Roush and Murphy (2007)

Disease	Pre-Vaccine Cases\n(per million/yr)	Case Reduction	Death Reduction
Diphtheria	158	100%	100%
Smallpox	250	100%	100%
Acute polio	141	100%	100%
Measles	3,044	99.99%	100%
Rubella	242	99.98%	100%
Pertussis	1,534	96.6%	99.7%
Hepatitis A	465	89%	88.7%
Acute hepatitis B	273	83.9%	83.6%
Pneumococcal	233	40.5%	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Vaccines reduced measles cases across US states



Key Milestones

1963

First measles vaccine developed by John Enders.

1971

MMR (measles-mumps-rubella) vaccine developed by Maurice Hilleman.

1980

Vaccination became mandatory for children entering kindergarten.

Before 1963, states had 100–1,000+ cases per 100,000. After the 1980 mandate, cases fell to near zero.

Smallpox: The Only Human Disease Eradicated Through Vaccination

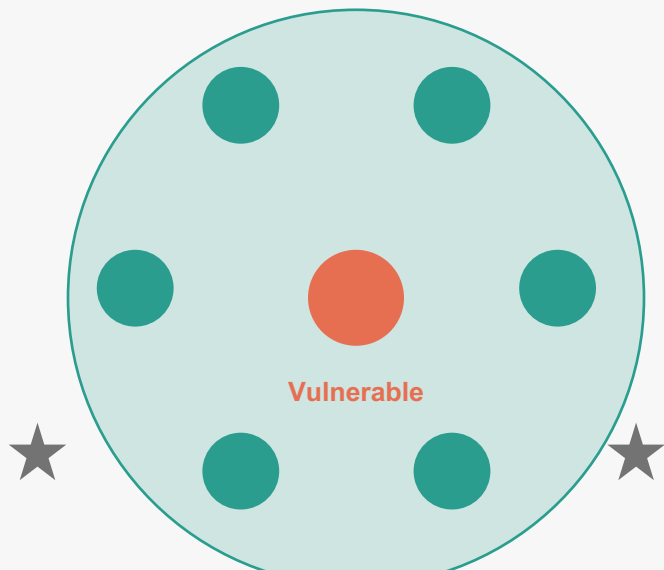
Why Smallpox Was Eradicated

- Virus infected only humans (no animal reservoir)
- Symptoms were easily distinguishable
- Vaccine was cheap and highly effective

Global Eradication Timeline

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- 1959** WHO passed initial resolution to eradicate smallpox globally
 - 1967** Intensified eradication campaign launched with more funding
 - 1977** Last natural case of smallpox occurred in Somalia
 - 1980** Smallpox officially declared eradicated worldwide

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Vaccinated Population Shields the Unvaccinated

Who Relies on Herd Immunity?

- Newborns who are too young to receive their first doses
- Immunocompromised individuals and those with weakened immune systems
- Cancer patients undergoing active chemotherapy
- Individuals with specific severe allergies to vaccine components

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

Case Reduction:

40.5%

Death Reduction: 31.3%

Hepatitis A

Case Reduction:

89.0%

Death Reduction: 88.7%

Acute Hepatitis B

Case Reduction:

83.9%

Death Reduction: 83.6%

Barriers to Complete Eradication

● Access Gaps

Inequitable global distribution prevents complete coverage.

● Supply Shortages

Intermittent availability puts children in developing regions at risk.

● Trust Challenges

Poor scientific understanding and misinformation reduce vaccine uptake.

Key Takeaways

Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- ✓ Vaccines achieved 100% elimination of multiple diseases in the US, including polio and smallpox.
- ✓ Development timelines have collapsed from over 100 years to under 1 year.
- ✓ Herd immunity protects entire communities, shielding those unable to be vaccinated.
- ✓ Coverage gaps persist for diseases like pneumococcal disease, hepatitis A, and hepatitis B.

Priority Action: Improve global access, expand coverage, and build trust in vaccination systems.